

Terminal & Cloud PDF Compilation Master Prompt & Reference Guide

Classification: DSOM Rule 11 & Rule 11.16 — Technical Book Design & Linux-Native PDF Compilation

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Purpose: Self-contained prompt to reproduce publication-grade PDFs from Markdown using the full Linux toolchain

Rule 11.16: Linux-Native PDF Compilation Mandate (WeasyPrint / Typst Engine)

When compiling technical documentation suites, forensic audit reports, or executive handbooks into publication-grade PDFs in Linux and WSL2 environments:

1. Zero Windows Host Dependency:

- The AI and compilation pipelines are strictly prohibited from depending on Windows host browser binaries (`chrome.exe` , `msedge.exe`) or Windows interop subprocesses (`cmd.exe /c start /wait` , `powershell.exe`) from within Linux/WSL2.
- Eliminates Windows profile lockouts, headless drawing stalls, interop command timeouts (code 124), and cross-filesystem path translation overhead.

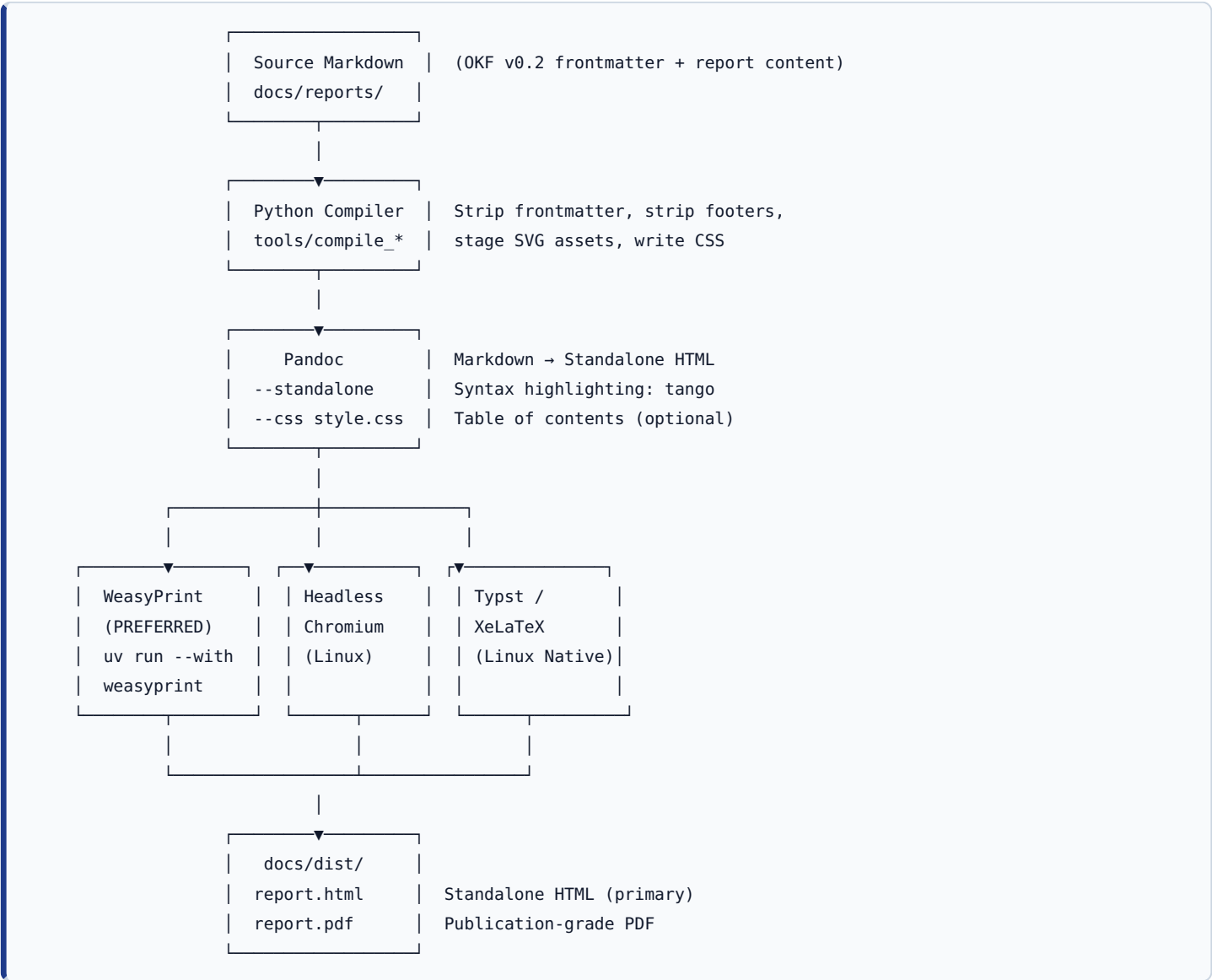
2. Linux-Native Headless PDF Engines (WeasyPrint First):

- The preferred, primary PDF compilation engine in Linux is native **WeasyPrint** executed via `uv` (`uv run --with weasyprint weasyprint <input.html> <output.pdf>`) or native headless Linux tools (`typst` , `xelatex`).
- The script must supply print-optimized CSS enforcing:
 - Pure white background (`#FFFFFF !important`).
 - Crisp, embedded standalone vector SVG diagrams (``).
 - Continuous multi-page table flow with repeated table headers (`thead { display: table-header-group; }` , `tr { page-break-inside: avoid; }`).
 - Standard A4 margins (`@page { size: A4 portrait; margin: 10mm 10mm 12mm 10mm; }`) and running footers (`Page X of Y`).

3. Deterministic Output & Disk Verification:

- The compiler script must synchronously assert that the output PDF exists on disk and has a non-zero byte size (`size > 10 KB`).
- If visual inspection is needed, verify pages locally using native Linux tools (`pdftoppm -png -r 150 <output.pdf> build/page`) without touching Windows.

1. Toolchain Overview



Required Software (Linux / WSL2)

Tool	Install Command	Purpose
Pandoc	<code>sudo apt install pandoc</code>	Markdown → Standalone HTML conversion
uv	<code>curl -LsSf https://astral.sh/uv/install.sh sh</code>	Python package runner (no virtualenv needed)
WeasyPrint	<code>uv run --with weasyprint weasyprint</code>	HTML → PDF (CSS @page support, running headers/ footers)
Python 3.10+	Pre-installed on most Linux	Compiler script execution
Chromium (optional)	<code>sudo apt install chromium-browser</code>	Alternative headless PDF renderer

2. The Terminal & Cloud CSS Framework (Pure White Print Mode)

This is the **complete, production-tested CSS** used in all our reports:

```

/* =====
Terminal & Cloud Design Framework – Print-Optimized Pure White
DSOM Rule 11 & Rule 11.16 | Zero Ink Waste | A4 Portrait | 10mm Margins
===== */

/* — Page Layout & Running Footer — */
@page {
  size: A4 portrait;
  margin: 10mm 10mm 12mm 10mm;
  @bottom-right {
    content: "Page " counter(page) " of " counter(pages);
    font-family: 'Inter', sans-serif;
    font-size: 8pt;
    color: #64748B;
  }
  @bottom-left {
    content: "PRIVATE AND CONFIDENTIAL (P&C) | Compile by: Harisfazillah Jamel";
    font-family: 'Inter', sans-serif;
    font-size: 8pt;
    color: #64748B;
  }
}

/* — Body Typography — */
body {
  font-family: 'Inter', -apple-system, BlinkMacSystemFont, "Segoe UI",
    Roboto, Helvetica, Arial, sans-serif;
  font-size: 8.8pt;
  line-height: 1.5;
  color: #0F172A; /* Dark Slate for maximum legibility */
  background-color: #FFFFFF !important; /* Pure white – ZERO toner waste */
  margin: 0;
  padding: 0;
}

/* Hide Pandoc's auto-generated title block (we use custom covers) */
#title-block-header {
  display: none !important;
}

/* — Heading Hierarchy — */
h1 {
  font-size: 17pt;
  font-weight: 800;
  color: #0F172A;
  border-bottom: 2px solid #0284C7; /* Sky Blue accent */
  padding-bottom: 6px;
  margin-top: 0;
  margin-bottom: 12px;
  page-break-after: avoid;
}

h2 {
  font-size: 12pt;
  font-weight: 700;
  color: #1E3A8A; /* Linux Blue */
  border-bottom: 1px solid #E2E8F0;
  padding-bottom: 4px;
}

```

```

    margin-top: 18px;
    margin-bottom: 10px;
    page-break-after: avoid;
}

h3 {
    font-size: 10pt;
    font-weight: 600;
    color: #0F172A;
    margin-top: 14px;
    margin-bottom: 6px;
    page-break-after: avoid;
}

/* — Paragraphs — */
p {
    margin-top: 0;
    margin-bottom: 8px;
}

/* — Blockquotes (Blue Info Callout) — */
blockquote {
    border-left: 3.5px solid #0284C7;
    background-color: #F8FAFC;
    color: #334155;
    padding: 8px 12px;
    margin: 10px 0;
    border-radius: 0 6px 6px 0;
    font-size: 8.4pt;
}

/* — Tables (Multi-Page Flow + Zebra Stripes) — */
table {
    width: 100%;
    border-collapse: collapse;
    margin: 10px 0;
    font-size: 7.6pt;
    page-break-inside: auto;           /* Allow tables to span pages */
}

thead {
    display: table-header-group;      /* REPEAT headers on every page */
}

tr {
    page-break-inside: avoid;         /* Never split a row across pages */
}

th {
    background-color: #F1F5F9;
    color: #0F172A;
    font-weight: 700;
    text-align: left;
    padding: 5px 7px;
    border: 1px solid #CBD5E1;
}

td {
    padding: 5px 7px;

```

```

        border: 1px solid #E2E8F0;
        vertical-align: top;
    }

tbody tr:nth-child(even) {
    background-color: #F8FAFC;           /* Light zebra striping */
}

/* — Code (Light Terminal Container) — */
code {
    font-family: 'JetBrains Mono', 'Fira Code', monospace;
    font-size: 7.8pt;
    background-color: #F1F5F9;
    color: #0F172A;
    padding: 1.5px 3.5px;
    border-radius: 4px;
    border: 1px solid #E2E8F0;
}

pre {
    background-color: #F8FAFC !important; /* Light gray — NOT dark/black */
    border: 1px solid #CBD5E1;
    border-left: 3.5px solid #1E3A8A;      /* Linux Blue left accent */
    border-radius: 6px;
    padding: 8px 10px;
    margin: 10px 0;
    overflow-x: auto;
    page-break-inside: avoid;
}

pre code {
    background-color: transparent !important;
    border: none;
    padding: 0;
    font-size: 7.4pt;
    color: #0F172A;
    line-height: 1.4;
    white-space: pre-wrap;                 /* Wrap long lines in PDF */
    word-break: break-all;
}

/* — Images & SVG Diagrams — */
img {
    max-width: 100%;
    height: auto;
    display: block;
    margin: 10px auto;
    page-break-inside: avoid;
    border: 1px solid #CBD5E1;
    border-radius: 6px;
    background-color: #FFFFFF;
}

/* — Horizontal Rules — */
hr {
    border: 0;
    height: 1px;
    background: #E2E8F0;
    margin: 16px 0;
}

```

```

}

/* — Callout Boxes (Warning / Note / Tip) — */
.callout-warning {
  background-color: #FEF2F2;          /* Light red pastel */
  border-left: 4px solid #EA580C;     /* Orange accent */
  padding: 12px 16px;
  margin: 10px 0;
  border-radius: 0 6px 6px 0;
  page-break-inside: avoid;
}

.callout-note {
  background-color: #F0F9FF;          /* Light blue pastel */
  border-left: 4px solid #0284C7;     /* Blue accent */
  padding: 12px 16px;
  margin: 10px 0;
  border-radius: 0 6px 6px 0;
  page-break-inside: avoid;
}

.callout-tip {
  background-color: #F0FDF4;          /* Light green pastel */
  border-left: 4px solid #16A34A;     /* Green accent */
  padding: 12px 16px;
  margin: 10px 0;
  border-radius: 0 6px 6px 0;
  page-break-inside: avoid;
}

/* — Card Accent Left Border Standard — */
.card-blue   { border-left: 5px solid #0284C7; background: #F8FAFC; padding: 10px; margin: 8px 0; }
.card-green  { border-left: 5px solid #16A34A; background: #F8FAFC; padding: 10px; margin: 8px 0; }
.card-amber  { border-left: 5px solid #CA8A04; background: #F8FAFC; padding: 10px; margin: 8px 0; }
.card-red    { border-left: 5px solid #DC2626; background: #F8FAFC; padding: 10px; margin: 8px 0; }
.card-purple { border-left: 5px solid #7C3AED; background: #F8FAFC; padding: 10px; margin: 8px 0; }

```

3. Checklist Before Every Compilation

- ☐ Source Markdown has NO raw `---` frontmatter (stripped by compiler)
- ☐ All SVG diagrams staged to both `build/` and `docs/dist/`
- ☐ CSS uses pure white `#FFFFFF` body background
- ☐ CSS uses light code containers `#F8FAFC` (NOT dark/black)
- ☐ Tables have `page-break-inside: auto` + `thead { display: table-header-group; }`
- ☐ No `<colgroup>` tags in HTML (strip them — they break table column widths)
- ☐ Running footer shows “Page X of Y” and report title
- ☐ Attribution: “Compile by: Harisfazillah Jamel”
- ☐ PDF file size > 10KB (assertion check)



Visual inspection: no blank pages, no orphaned headings, no clipped tables